VF Asia Footwear Laboratory Testing Manual

Test Method for DIN Abrasion	**Test method # VF- C01** **REF #: TM-R- 002** **Page 1 of 3**
Date Issued: July, 2017 **Revision #: 1** **Reason for revisions: 1. Reference # was changed from DIN 53516 to DIN 4649.** **2. Indicate pre-buffing of test specimen.**	**Created by: Nila Babia Erazo**

PURPOSE AND SCOPE: To measure the resistance of rubber, thermoplastic rubber, and plastic to abrasion, by traversing a loaded, rotating test sample over an abrasive cloth. The abrasive cloth is attached to a cylinder that is rotating perpendicular to the direction of the test sample.

REFERENCE: DIN ISO 4649: 2006-11

EQUIPMENT AND MATERIAL:

1. Abrasion machine [A], meeting the requirements of DIN 4649.
2. Drilling die, 16.0 +/-0.2 mm in diameter.
3. Electric drill mounted in a drill stand. (The stand must allow the drilling die to move up and down in a perfectly vertical manner.)
4. Standard elastomer [A], meeting the requirements of DIN 4649.
5. Cyanoacrylate (super-glue), suitable for bonding rubber and plastics.
6. Corundum abrasive cloth [A] of grain size 60 bonded to cotton twill and meeting the requirements of DIN 53516.
7. Aluminum disc, diameter of 16.0+/-0.2 mm, and minimum thickness of 6mm.
8. Balance, capable of weighing to the nearest 0.0001 grams.
9. Soft bristle brush.
10. Shore A durometer.
11. Rubber, thermoplastic rubber, and plastic (typically outsole materials).

SAMPLE PREPARATION:

1. Condition the sample for at least 24 hours at the standard laboratory environment at 23.0±2.0 ˚C temperature and 55 ±10 % Relative Humidity.

2. Using the drilling die, cut three (3) test samples have a diameter of 16.0+/- 0.2 mm, from varying locations on the test material. The test material must have a minimum thickness of 6mm. Test samples must have parallel top and bottom surfaces, and sides which are perpendicular to the top and bottom surfaces (i.e., cylindrical in shape).

Note: *If a sample with a thickness of less than 6 mm must be used, it should be cut in the manner described above. The test sample must then be bonded to a second piece of rubber which is also cut in the same manner and has a minimum durometer of 65 Shore A. The two pieces must have a combined thickness of at least 6mm. Cyanoacrylate ("super-glue") is the recommended adhesive.*

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EQUIPMENT PREPARATION:

After every ten (10) test runs, the "standard elastomer" test sample must be evaluated to ensure that the equipment is properly calibrated. The testing of the "standard elastomer" is to be performed using the same test procedure which is described below. One piece of the "standard" elastomer is to be measured three (3) times, and averaged.

The equipment is properly calibrated when the "standard elastomer" produces an average weight loss of 200+/-20 mg. If the weight loss is more than 220 mg, the sandpaper is too aggressive and must be adjusted. This can be done by inserting an aluminum disc into the sample holder. Pass the aluminum disc over the sandpaper until testing with the standard elastomer produces a weight loss of 220 mg or less. If the weight loss is less than 180 mg, the sandpaper is not aggressive enough and must be replaced.

Notes:

1. *If any measurement varies by more than 15 mg from the others, it should be discarded and another measurement should be taken.*
2. *Since the surface of the rubber has wax, oil or contaminants, it is a must to pre-buff the specimen before the test by allowing it to rotate and traverse at least half of the length of the drum.*
3. *Clean the sand paper in every after test with soft bristle brush, to remove any debris which may be present from previous testing. The use of vacuum cleaner is also acceptable. (Brushes with metal bristles are not recommended as the life of the abrasive sheet will be shortened)*

REPETITIONS: Test three (3) samples per material.

TEST PROCEDURE:

Test should be conducted at controlled laboratory environment at 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

1. Determine the specific gravity (**SG**) of the test material according to **VF Test Method: VF – A04**
2. Load the sample holder with the 250 gram and 500 gram weights (bringing the total weight of the sample holder to 1 kg, as the sample holder weighs 250 grams).
3. Pre-buff the test sample before the test by allowing it to rotate and traverse at least half of the length of the drum.

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4. Remove any burrs, and then brush any dust or loose particles from the sample with a soft bristle brush.
5. Weigh the test sample to the nearest 0.001 gram (**W_1**).
6. Using a soft bristle brush, remove any debris from the sandpaper which may be present from previous testing. The use of vacuum cleaner is also acceptable. (*Brushes with metal bristles are not recommended as the life of the abrasive sheet will be shortened)*
7. Insert the test sample in the sample holder of the DIN abrader such that the surface is flush with the end of the retaining rim. Turn the knurled nut two complete rotations in the counter-clockwise direction. This should result in the test sample protruding 2.0 +/-0.2 mm beyond the end of the retaining rim. (For test sample which may have more than 2 mm abraded, the test must be stopped at the halfway point. The test sample should be adjusted so that it once again protrudes 2 mm beyond the retaining rim, and then testing can be resumed).

Note**:** *If using Hampden test equipment, run the test without the counter balance.*

8. Lower the sample holder to the starting position, turn on the machine and vacuum, and allow the sample to rotate and traverse the entire length of the drum (40 meters/pass).
9. At the completion of testing, remove the test sample from the holder, remove any burrs with your fingers, and brush any dust or loose particles from the sample with a soft bristle brush.
10. Re-weigh the test sample to the nearest 0.001 gram (**W_2**).

CALCULATION:

$$\text{DIN Abrasion (volume loss in mm}^3\text{)} = \frac{(W1-W2)}{SG} \times \frac{200mg}{S_A} \times 1000$$

S_A= Actual reading of the standard elastomer

REPORT: The average of the three test samples, to the nearest one (1) mm^3.